reduced the number of predatory fish to such an extent that the number of herbivorous crabs has expanded dramatically. The resulting increase in the consumption of marsh vegetation has had a knock-on effect on other species that depend on it.

Trophic cascades can also be disturbed by the introduction and spread of a nonnative species, as happened when the omnivorous mud crab indigenous to waters on the east coast of North America and Mexico became common in the Baltic Sea in the 1990s. Crabs, which are keystone species in many coastal food webs, feed on benthic (seafloor) communities —bivalves, gastropods, and other small invertebrates—with devastating efficiency, creating a strong top-down cascade. The increase in the number of mud crabs in the Baltic, where there are no equivalent predators, resulted in a dramatic decline in the mix of benthic invertebrate species. This in turn led to an increase in floating nutrients, which ultimately boosted phytoplankton rather than species on the seafloor. The net effect of the crabs' arrival was to transfer nutrients from the seafloor to the water column —the water between the sediment and the surface—and to degrade the ecosystem.

Bottom-up cascades

If a plant—a primary producer—is removed from an ecosystem, a bottom-up cascade may result. For example, if a fungal disease causes grass to die-off, a rabbit population that depends on it will crash. In turn, the predators that eat rabbits will starve or be forced to move away, and the entire ecosystem could break down. Conversely, if planting or conservation efforts boost the mix of plant life, more herbivores (including the pollinators that help plants reproduce and spread) will be attracted, and with them more predators.

In the bottom-up model, the responses of herbivores and their predators to increased plant variety follow in the same direction: more plants support